

Lab 1 – Ideal Gas Law

CHE 260 – Thermodynamics and Heat Transfer

Group Members

Chance Hattrick	1011852434
Ammar Hasham	1011941124
Muhammad Hasnain Heryani	1011884695

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1 Introduction

1.1 Ideal Gas Law

Gases provide a fundamental foundation for understanding chemistry, and the quantitative laws governing their behavior establish a direct connection between macroscopic and particle-level properties. A principal example of this relationship is the Ideal Gas Law. The law was formulated in 1834 by Benoît Paul Émile Clapeyron, who combined empirical relationships including Boyle’s Law and Charles’s Law [1]. The most common formulation is:

$$PV = NR_uT \quad (1)$$

where P is the absolute pressure of the system, V is the volume, N is the number of moles of gas, R_u is the universal gas constant (8.314 J/(mol · K)), and T is the absolute temperature expressed in Kelvin.

For this experiment, the absolute pressure is calculated using:

$$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}} \quad (2)$$

where P_{gauge} is the pressure measured by the gauge and P_{atm} is the atmospheric pressure measured in the laboratory.

Since the number of moles is equal to the total mass of the gas divided by its molar mass, the Ideal Gas Law can be rearranged

as:

$$PV = mRT \quad (3)$$

where m is the mass of the gas and R is the specific gas constant, obtained by dividing R_u by the molar mass. For air, this value is approximately 287 J/(kg · K) [2].

This formulation is particularly important for this experiment because the mass of air introduced into the tank can be calculated from the measured mass flow rate.

Under the ideal gas model, particles are assumed to move randomly and independently, with negligible molecular volume and negligible intermolecular forces except during collisions. These collisions are assumed to be perfectly elastic, conserving both momentum and kinetic energy [3].

1.2 Thermodynamic State

Properties and Path Independence

To properly understand the Ideal Gas Law, it is also important to understand thermody-

dynamic properties, particularly state properties. A thermodynamic property is a measurable or calculable macroscopic characteristic of a system whose value can be determined without knowing the previous history of the system [4].

These properties describe the thermodynamic state of a system and include pressure, temperature, volume, mass, and density. Their values depend only on the current state of the system rather than the path taken to reach that state. This characteristic is known as path independence.

This concept is especially important for this laboratory experiment, as the first part compares two different methods of equalizing pressure between interconnected tanks of air.

In the first method, air is transferred rapidly through a solenoid valve, while in the second method, air is transferred gradually through a needle valve. Although the pressure and temperature histories may differ between these processes, both methods are expected to produce approximately the same final equilibrium conditions, provided the initial conditions and final constraints are equivalent.

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Path independence can also be demonstrated through the change in internal energy of a system:

$$\Delta U = U_f - U_i \quad (4)$$

where U_f and U_i represent the final and initial internal energies, respectively. Since internal energy is a state property, its change depends only on the initial and final thermodynamic states.

This differs from heat transfer and work, which are path-dependent quantities.

1.3 Heat Transfer and the First Law of Thermodynamics

An important consideration when studying gas expansion is the transfer of energy between a system and its surroundings. This relationship is described by the First Law of Thermodynamics, which states that energy is conserved [4].

For a closed system, neglecting changes in kinetic and potential energy, the First Law can be written as:

$$\Delta U = Q - W \quad (5)$$

where ΔU is the change in internal energy, Q is the net heat transferred into the system, and W is the work done by the system.

In this experiment, the two interconnected rigid tanks can be treated as a single closed system. Since the total volume remains constant, there is no boundary work. Assuming negligible other work interactions, the First Law becomes:

$$\Delta U = Q \quad (6)$$

Heat transfer is particularly relevant when comparing the rapid and gradual pressure-equalization methods. During rapid expansion through the solenoid valve, air moves between the tanks over a relatively short period, limiting the time available for heat exchange with the surroundings.

Consequently, temporary temperature differences may develop between the two tanks.

In contrast, gradual expansion through the needle valve provides more time for heat exchange with the surrounding environment, allowing the system to remain closer to thermal equilibrium throughout the

process.

Although the two methods may produce different temperature histories, they are expected to approach similar final equilibrium conditions.

1.4 Air Compressibility and Intermolecular Forces

Finally, another important characteristic of gases is their compressibility, which describes how their volume or density changes in response to variations in pressure.

Unlike liquids and solids, gases are highly compressible due to the relatively large distances between their molecules [5].

In this experiment, the two tanks are rigid, meaning their volumes remain constant throughout the process. Consequently, changes in pressure are associated with changes in the mass, density, and temperature of the air within each tank.

This relationship can be described using the Ideal Gas Law, rearranged as:

$$\rho = \frac{P}{RT} \quad (7)$$

where ρ represents the density of air.

However, the Ideal Gas Law assumes

that the volume occupied by individual gas molecules and the intermolecular forces between them are negligible.

In reality, attractive and repulsive intermolecular forces influence gas behavior, particularly at high pressures or low temperatures. As a result, real gases may deviate from the predictions of the Ideal Gas Law.

These deviations can be quantified using the compressibility factor, Z :

$$Z = \frac{Pv}{RT} \quad (8)$$

where v represents the specific volume of the gas.

For an ideal gas, $Z = 1$, while values different from unity indicate deviations from ideal gas behavior.

In this experiment, the maximum gauge pressure was approximately 40 psi, corresponding to an absolute pressure of approximately 3.8 bar.

Under these relatively moderate conditions, air is expected to behave approximately as an ideal gas, with a compressibility factor close to unity.

Therefore, although intermolecular forces may introduce some uncertainty into

the calculated tank volumes, their effects are expected to be relatively small.

2 Results and Discussion

2.1 Part 1: Path Independence and Tank Volume Ratio

Part 1 of the lab discusses the comparison between the movement of air between two identical volume tanks, one with a gauge pressure of 40 PSI to another with a gauge pressure of 0-PSI. The main point of comparison is the differences in a slow equalization of pressure versus a rapid equalization of pressure.

2.1.1 Processed Results: Rapid and Slow Expansion

Figures 1–2 show the measured pressure and temperature histories for the rapid and slow expansions.

Taking averages across the marked time ranges on Figures 1–2 for final pressure and temperature values, we obtain the following values:

$$P_{2R} = 24.13 \text{ PSI}, P_{2L} = 24.03 \text{ PSI}$$

$$T_{2R} = 28.33^\circ\text{C}, T_{2L} = 25.25^\circ\text{C}$$

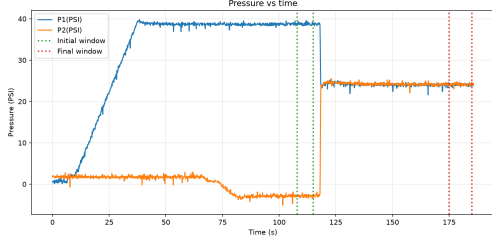


Figure 1: Tank pressures during the Part 1a rapid expansion.

Using these values with Ideal Gas Law, we can determine the proportionality of pressure, volume and temperature between the two tanks at the final state. It can be assumed that the ratio of the final pressure, volume and temperature of the left tank is equal to the final pressure, volume and temperature of the right tank since both tanks are in equilibrium with each other at the final values. The following calculations show the resultant volume ratio of the rapid equilization of pressure

$$\begin{aligned}\frac{P_{2L}V_L}{T_{2L}} &= \frac{P_{2R}V_R}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{P_{2R}}{P_{2L}} \cdot \frac{T_{2L}}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{24.13}{24.03} \cdot \frac{25.25}{28.33} \\ \frac{V_L}{V_R} &= 0.895\end{aligned}$$

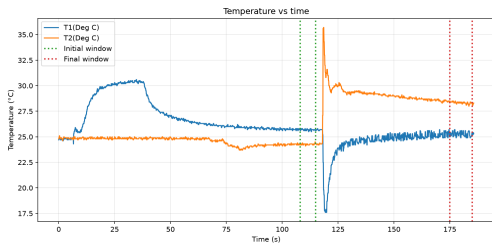


Figure 2: Tank temperatures during the Part 1a rapid expansion.

Similar to Figures 1–2, the following Figures 3–4 show the pressures and temperatures of both the left and right tank during the slow equilization of pressure part of the lab.

Taking averages of the final pressures and temperatures at the marked time ranges on the graphs, similar to Figures 1 and 2, we

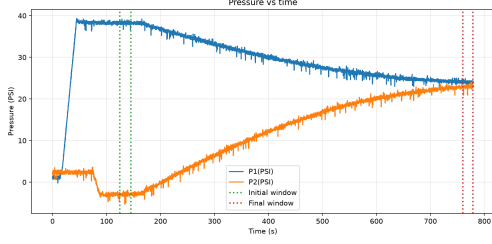


Figure 3: Tank pressures during the Part 1b slow expansion.

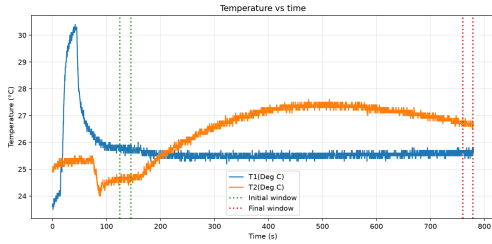


Figure 4: Tank temperatures during the Part 1b slow expansion.

obtain the following values:

$$P_{2R} = 22.94 \text{ PSI}, P_{2L} = 24.04 \text{ PSI}$$

$$T_{2R} = 26.70^\circ\text{C}, T_{2L} = 25.60^\circ\text{C}$$

Then applying the Ideal Gas Law in the same way, we obtain the following:

$$\begin{aligned} \frac{V_L}{V_R} &= \frac{P_{2R}}{P_{2L}} \cdot \frac{T_{2L}}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{22.94}{24.04} \cdot \frac{25.60}{26.70} \\ \frac{V_L}{V_R} &= 0.915 \end{aligned}$$

2.1.2 Heat Transfer and Net Heat Transfer

Though heat transfer is a path-dependent quantity, the net heat transfer in both expansion cases is roughly the same because the initial and final thermodynamic states are approximately identical. However, the heat transfer histories differ significantly between the two processes.

During Part 1A, rapid expansion, once the solenoid valve is opened, the temperature in the left tank experiences a sharp decrease while the temperature of the right tank suddenly increases, before both temperatures slowly reach thermal equilibrium. This sharp change in temperature occurs because the flow of air between the tanks is faster than the rate at which heat can be exchanged with the surroundings.

Meanwhile, in the slow expansion experiment, once the needle valve is opened, the temperatures of the two tanks gradually reach thermal equilibrium over a much longer period of time, without sudden changes in temperature.

When determining the net heat transfer, both tanks can be considered as a single

closed system. Because the air initially and finally reaches approximately ambient temperature, there is no boundary work, and the total mass of the system remains constant, the First Law of Thermodynamics gives:

$$\Delta U = Q - W \quad (9)$$

Since the tanks are rigid, no boundary work is performed, and assuming negligible other work interactions:

$$W = 0 \quad (10)$$

Additionally, since the initial and final temperatures are roughly equal, the change in internal energy is approximately zero:

$$\Delta U \approx 0 \quad (11)$$

Therefore, the net heat transfer is:

$$\boxed{Q_{\text{net}} \approx 0} \quad (12)$$

2.1.3 Path Independence of State Properties

As mentioned previously, a thermodynamic state property is a property whose value depends only on the current state of the system and not on the process used to reach that state. The experiment demonstrates the path

independence of state properties by comparing two different methods of pressure equalization between interconnected tanks.

In Part 1A, air is transferred rapidly through a solenoid valve, leading to sudden changes in pressure and temperature, as seen in Figures 1–2. Alternatively, in Part 1B, air is transferred gradually through a needle valve, producing slower changes in these properties, as seen in Figures 3–4.

Despite these differences in the paths of the two processes, the final equilibrium pressure is approximately the same when starting from equivalent initial conditions. Therefore, the experiment illustrates how different processes can connect the same initial and final equilibrium states, while thermodynamic state properties remain independent of the path taken between them.

2.2 Part 2: Initial Mass and Volume of the Left Tank

2.2.1 Processed Pressure, Temperature, and Mass Flow Results

Figures 5–6 show the measured tank pressures and temperatures during Part 2.

2.2.2 Initial Mass and Tank Volume

Figure 7 shows the measured mass flow rate used in the added-mass calculation.

Knowing that all added mass is measured by the sensor, we can calculate the added mass in the left tank as the integral of the mass flow rate - time graph in Figure 7. Using numerical methods, we can find that the total mass inside the tank is 23.259g.

Since the mass inside the system after any time is known, we can create a system of equations to calculate V and initial mass.

$$PV = (m_i + \delta m)RT$$

$$P(30) = 17.6$$

2.2.3 Sources of Error and Limitations

There are several key sources of errors and limitations on this section of the lab, either relating to random human error or systematic error with the apparatus.

When setting the pressure of the system to 40 PSI, we were reliant on a teammate manually checking the gauge pressure. This is limited by human reaction time and uncertainty in the gauge pressure reading, both due to parallax error and uncertainty in measurements. Thus, there was a lot of error with

setting the initial pressure to exactly 40 PSI.

This is seen through the sensor data which showed that the pressure was slightly below our target of 40 PSI.

Additionally, sensor data displayed that the mass flow rate was not completely constant at 50 g/s, even when temperature and pressure have stabilized for the air in the tank.

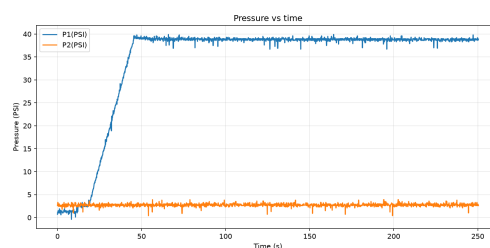


Figure 5: Tank pressures during Part 2.

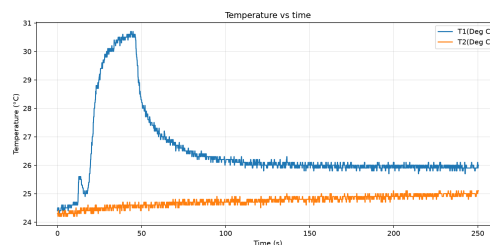


Figure 6: Tank temperatures during Part 2.

2.2.4 Effect of Air Compressibility

The compressibility of air can greatly affect experimental results of pressure and volume, especially as experimental conditions diverge

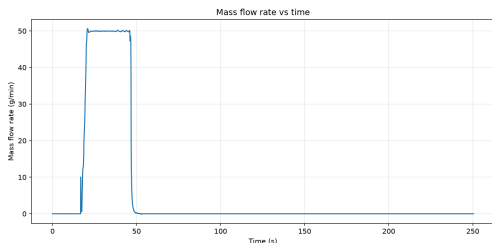


Figure 7: Mass flow rate during Part 2.

from assumptions made by the ideal gas equation. The ideal gas equation assumes that temperatures are kept high and pressures are kept low. When pressures are kept low, intermolecular forces can be assumed to be negligible. With this model, the compressibility of air can be modelled to be constant. As the pressure increases, electromagnetic repulsions and attractions between particles greatly contribute to the behavior of the gas. This causes the linear model of air compressibility to break down, displaying the inaccuracies in the ideal gas model. Since the intermolecular forces increase as pressure increases, our calculated compressibility is much higher than the actual compressibility.

One can model this using the compressibility factor, where

$$Z = \frac{Pv}{RT}$$

$$\text{or } PV = ZRT.$$

Under constant volume, this measures the divergence of a physical model from Gay-Lussac's Law.

During our experimentation, the maximum gauge pressure found was around 40 psi. Given that this is a similar gauge pressure to that seen in pressurized car tire, this is high enough where our apparatus diverges from assumptions made by the ideal gas model and we should check the compressibility factor during our process.

If we hold the volume V constant, we can calculate the compressibility factors at low and high pressures.

At $t = 10$:

$$1.3 \cdot v = Z_1 \cdot R \cdot 317.85,$$

$$Z_1 = \frac{1.3 \cdot v}{317.85 \cdot R} = \frac{2}{489} \cdot \frac{v}{R}.$$

At $t = 50$:

$$52.2 \cdot v = Z_2 \cdot R \cdot 301.45,$$

$$Z_1 = \frac{52.2 \cdot v}{301.45 \cdot R} = \frac{1050}{6029} \cdot \frac{v}{R}.$$

Thus, through these calculations, we can see that the compressibility of air significantly increases at higher pressures.

3 Conclusion

References